Agentic AI in mathematical research

A brief disclaimer

I am not an expert in this field. I have been exploring it out of curiosity and to stay up to date.

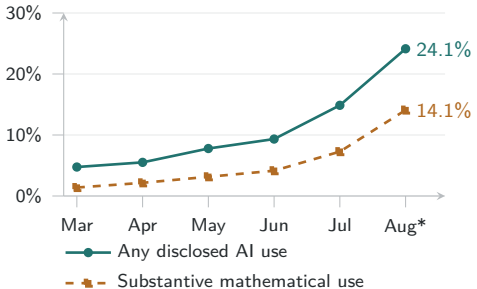
This talk neither advocates nor argues against using AI to write papers.

- A brief history and overview of mathematical research with LLMs, focusing on agentic approaches.
- What the literature tells us about successful approaches and their limitations.
- A case study where I build an AI research team to write a paper.

AI use and mathematical output

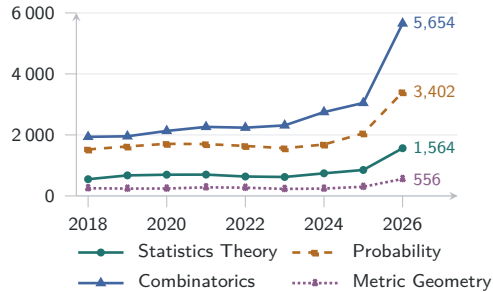
AI disclosures on arXiv

Share of math-related submissions, 2026



Submissions by field

New preprints, January–August each year

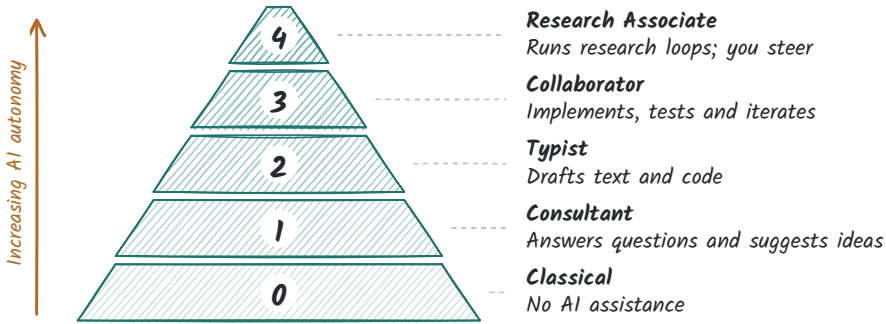


Left: 3,575 of 32,944 preprints disclosed AI use. *August covers 1–20.
Right: primary-category counts. Growth alone does not identify an AI effect.

Jin, Ke & Sui (2026), Table 1 **Explore arXiv Math Pulse** GitHub source.

How have you used AI in research?

Which is the highest level you have used?



Different tasks in one project may use different levels.

Adapted from Zimmer et al. (2026), *The Agentic Researcher*, Section 2.

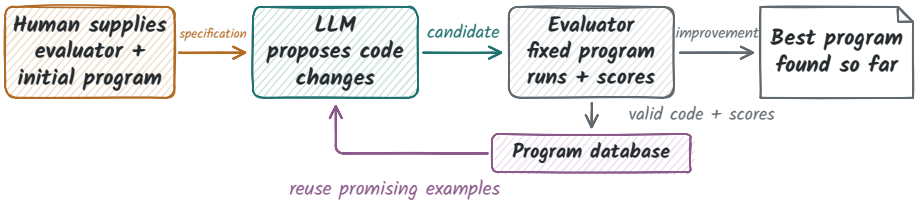
A short history of math research with LLMs

From language models to research agents

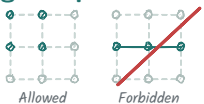
A brief history, with a focus on mathematics.

FunSearch Romera-Paredes et al., *Nature* (2024)

Paper recap: *Mathematical discoveries from program search with large language models*



Larger cap sets

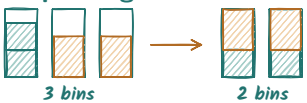


496 → 512

points in 8 dimensions

No three distinct points on a line in $\mathbb{F}_3^n$.
A new construction, not a proof of maximality.

Better bin-packing heuristics

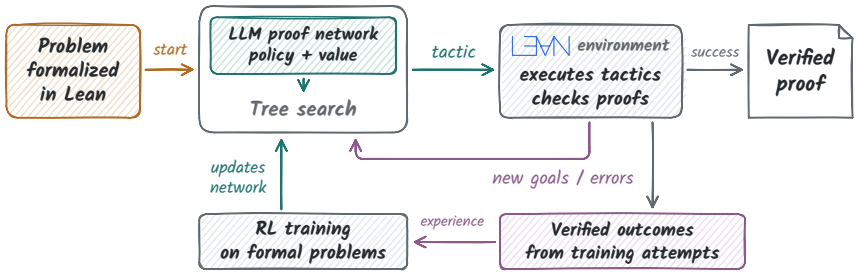


Pack arriving items into as few fixed-capacity bins as possible.

Workflow adapted from Fig. 1; results: Fig. 4 and Table 1. Romera-Paredes et al., *Nature* **625**, 468–475 (2024).

AlphaProof Hubert et al., *Nature* (2026)

Paper recap: *Olympiad-level formal mathematical reasoning with reinforcement learning*



Search: the LLM proposes tactics; Lean returns new proof states.

Learning: RL updates the network. Hard targets get further training on related variants.

IMO 2024: silver-medal level — 28/42 points with AlphaGeometry 2.

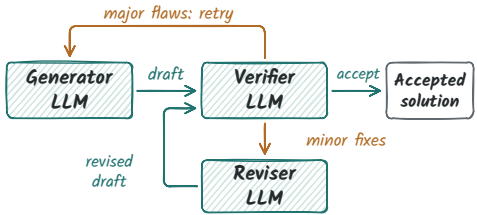
AlphaProof solved 3 problems; AlphaGeometry 2 solved 1.

Adapted from Figs. 1–2 and the AlphaProof section. *Nature* **651**, 607–613 (2026).

Aletheia Feng et al., arXiv (2026)

Paper recap: *Towards Autonomous Mathematics Research*

Gemini Deep Think agents



LLM judgment; no formal guarantee (Fig. 1).

Reported mathematical results

- Derived new formulas for eigenweights.
- Erdős: 2 full + 2 partial new solutions from 700 problems, assessed by experts.

Other LLM research systems

The AI Scientist *Nature* (2026)
Ran ML experiments and wrote a paper that passed a workshop's first review round.

Google Co-Scientist *Nature* (2026)
Agents proposed and refined leukaemia drug combinations, validated in laboratory experiments.

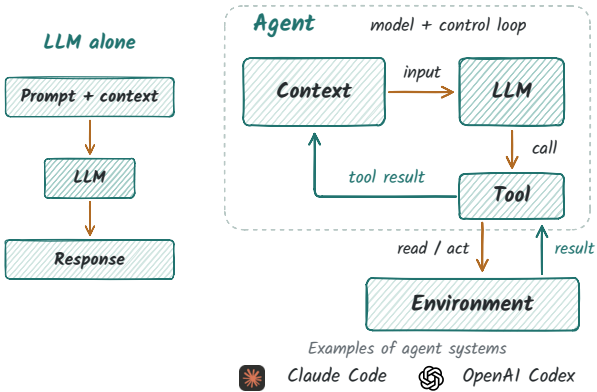
Virtual Lab *Nature* (2025)
LLM agents designed SARS-CoV-2 nanobodies with binding confirmed in laboratory tests.

OpenAI: Navier–Stokes report (2026)
Reported blow-up under smooth forcing, formalized in Lean. Parallel agent groups explored different approaches; researchers redirected effort and shared promising intermediate results.

Agentic systems

From LLMs to research teams

Environment, context, tools and agents



Environment
The workspace an agent can access: files, data, code and external services.

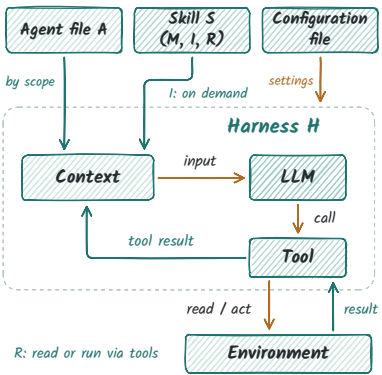
Context
The information given to the LLM now:
instructions, messages and tool results.

Tool
An interface to observe or change the environment, e.g. search or Python.

Agent
An LLM-based system that uses tools and chooses its next step from results.

Based on OpenAI's Codex harness documentation and Claude Code's agentic loop.

Skills, agent files and the harness



Skill $S = (M, I, R)$
M — **metadata**: name and task relevance.
I — **instructions**: method, decision rules, checks and expected output.
R — **resources**: optional documents, code, examples and templates.

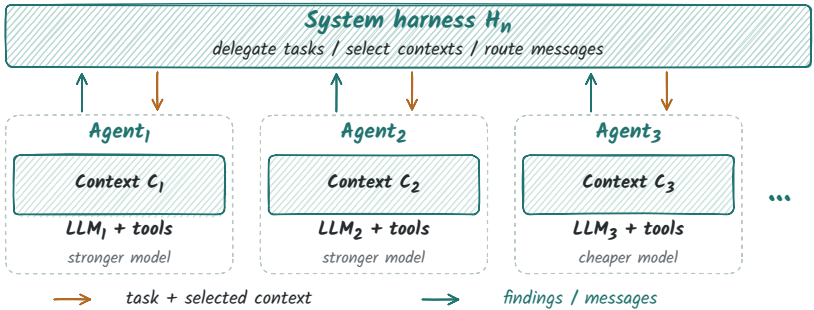
Agent instruction file $A = (D, I_A)$
D: scope; I_A: standing instructions.
E.g. `AGENTS.md`: project guidance loaded into the context within its scope.

Harness H
Orchestration software. A configuration file sets roles, models, tools, skills and routing.

Working abstractions for this talk. Implementation: skills, `AGENTS.md`, Codex harness (OpenAI Docs).
Foundations: ReAct — Yao et al. (2022), Voyager — Wang et al. (2023), SWE-agent — Yang et al. (2024).

Multi-agent systems: who sees what?

Context management — the "holy grail": the right context for each task.



H_n is the software coordinating n agents, their contexts and communication.

A configuration file sets roles/models/tools; skills and agent files supply instructions.

Same or different models. Sources: AutoGen (Wu et al., 2023); OpenAI agent configuration.

Why manage context and communication?

1. Keep context focused Liu et al. (2024); Modarressi et al. (2025)
Give each agent the information it needs. Long conversations can make relevant details harder to find and use.

2. Give review a fresh context Sharma et al. (2023); Dhuliawala et al. (2023)
Review the draft and criteria without the writer's history. Framing can bias feedback; independent verification can help.

3. Match the model to the task Ong et al., *RouteLLM* (2024)
Routing between stronger and cheaper models can improve the balance of cost and quality.

4. Coordinate selectively Du et al. (2023); Wang et al. (2025)
Let agents share useful findings. Removing agents or messages that repeat the same work can save time and improve results.

5. Explore alternative proofs in parallel Xin et al., *DeepSeek-Prover-V1.5* (2024)
Agents with different contexts can sample and check alternative proofs simultaneously. More attempts may find a solution, at greater total compute cost.

Effects depend on the task and models. Fresh-context review is a design inference from bias and verification studies.

Five heuristics for agentic research

Heuristic	Intuition
Ask for a proof or a counterexample. [1]	Let the agent show that the claim is false; do not let it silently change the problem to obtain a proof.
Delegate precise mathematical obligations. [2]	Explicit assumptions and dependencies make partial progress and remaining gaps visible.
Explore distinct approaches independently. [3]	Different starting points can reveal approaches that others miss.
Preserve each claim's status and evidence. [4]	Reuse must not turn a conjecture or rejected argument into an "established lemma".
Fix identified errors; stop if no progress is made. [5]	Check that each error was repaired. Repeated approval is not evidence of a correct proof.

Evidence-informed design heuristics: support ranges from benchmark experiments to research-system case studies.

[1] Petrov et al. (2025); [2] Zhao et al. (2026); [3] Zhu et al. (2026);
[4] Xu et al. (2026); [5] Zheng et al. (2026).

Agentic research: a worked example

What do we need?

- An **agent system**: in my case, Codex.
- A precise **mathematical problem statement**.
- Relevant **related work**: definitions, known results and useful methods.
- `instruction.md`: the task, constraints and expected output.
- `skills.md`: an index of reusable methods and checks.
- `AGENTS.md`: standing project instructions for the agents.

Here, `instruction.md` and `skills.md` are our project documents.
Each actual Codex skill has its own `SKILL.md`.

Step 1: define the mathematical objects

A **simple point process** X on $\mathbb{R}^d$ is a random set of distinct points, finite in bounded regions. Its **count** in B is

$$X(B) = \#(X \cap B).$$

The **factorial moment measure** counts distinct, ordered n-tuples:

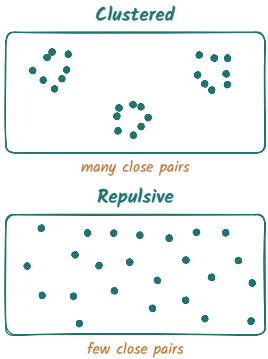
$$\alpha^{(n)}(B_1 \times \cdots \times B_n) = \mathbb{E} \sum_{(x_1,\ldots,x_n) \in X^n_{\neq}} \prod_{i=1}^{n} \mathbf{1}_{B_i}(x_i).$$

At order two,

$$\alpha^{(2)}(B_1 \times B_2) = \mathbb{E}[X(B_1)X(B_2)] - \mathbb{E}[X(B_1 \cap B_2)].$$

Interpretation: the expected number of ordered pairs of distinct points, with the first in B_1 and the second in B_2.
A point is never paired with itself.

Waagepetersen (2022), slides 7–13; Last and Penrose, Section 4.4.



Schematic point patterns.

Product densities and correlations

When $\alpha^{(n)}$ has a locally integrable density, the **product density** is defined by

$$\alpha^{(n)}(A) = \int_A \rho^{(n)}(x_1, \ldots, x_n)\, \mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

At first order, $\rho^{(1)}$ is the density (derivative) of the **mean count** $\alpha^{(1)}(B) = \mathbb{E}[X(B)]$:

$$\rho^{(1)}(x) = \frac{\mathrm{d}\alpha^{(1)}}{\mathrm{d}x}(x), \qquad \rho^{(2)}(x, y) = \frac{\mathbb{E}[X(\mathrm{d}x)X(\mathrm{d}y)]}{\mathrm{d}x\, \mathrm{d}y} \quad (x \neq y).$$

With positive intensity densities $\rho^{(1)}(x_i)$, define the ***n*-point correlation function**:

$$g_n(x_1, \ldots, x_n) = \frac{\rho^{(n)}(x_1, \ldots, x_n)}{\prod_{i=1}^n \rho^{(1)}(x_i)}.$$

Stationarity: shifting all points leaves their joint law unchanged: $X + t \overset{d}{=} X$ for every $t \in \mathbb{R}^d$. Then $\rho^{(1)}(x) = \lambda$ is constant and $g_n = \rho^{(n)}/\lambda^n$ (assume $0 < \lambda < \infty$).

Relative to Poisson with the same intensity:

$g_n > 1$**: more such tuples.** $g_n < 1$**: fewer such tuples.**

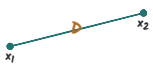
For pairs: clustering or repulsion at the specified locations and scale. $g_2 = 1$ alone does not imply Poisson.

Waagepetersen (2022), slides 9–13; Last and Penrose, Section 4.4.

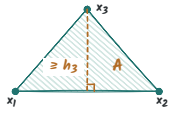
Step 1: the precise problem

Fix $M, \lambda_0 > 0$ and choose $0 < h_3 < \frac{\sqrt{3}}{2}M$, $0 < h_4 < \sqrt{\frac{2}{3}}M$.

Find **one stationary, simple** point process X on $\mathbb{R}^3$ with intensity λ_0, finite local moments through order four and product densities.



Distance D



Area A



Volume V

For some constants $C_2, C_3, C_4 > 0$, require **simultaneously**:

$$g_2(x_1, x_2) = \frac{C_2}{D}, \qquad g_3(x_1, x_2, x_3) = \frac{C_3}{A}, \qquad g_4(x_1, x_2, x_3, x_4) = \frac{C_4}{V}.$$

Almost everywhere on nondegenerate configurations of diameter $\leq M$, with every triangle altitude $\geq h_3$ and every tetrahedron altitude $\geq h_4$.

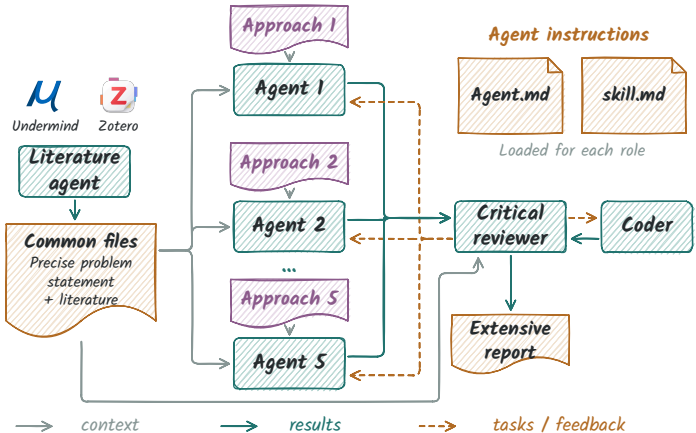
C_2, C_3, C_4 are chosen jointly with the process, not prescribed independently.

Current construction: sufficiently small λ_0; arbitrary prescribed intensity remains open.

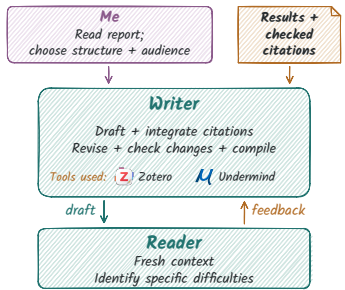
Configuration file, agent files and execution

Research approaches

1. Poisson cluster process
2. Cox process
3. Hawkes process
4. An approach of the agent's choice
5. Prove non-existence



Step 3: Writing the paper



Complete-task statistics

Measure	Complete task
Input tokens	131,866,681
Cached input (included above)	128,329,472
Uncached input	3,537,209
Output (includes reasoning)	666,705
Reasoning (subset of output)	286,459
Total tokens	**132,533,386**
Task-turn runtime	**3:47:06**
Research coordinator runtime	**3:36:43**
API-equivalent cost	**$78.81**

Writing stage: token use and cost

Participant	Uncached input	Cached input	Output/reasoning	Cost
Coordinating writer	0.30M	3.5M	0.15M	$5.60
Simulation specialist	0.15M	1.4M	0.10M	$3.16
Fresh reader	0.06M	0.1M	0.02M	$0.68
Total	**0.51M**	**5.0M**	**0.27M**	**$9.44**

Results and personal feedback

Poisson cluster model

Homogeneous Poisson parents Π each generate **0–3 children**, with jointly chosen offsets C_z:

$$X = \bigcup_{z \in \Pi} \big(\{z\} \cup (z + C_z) \big).$$

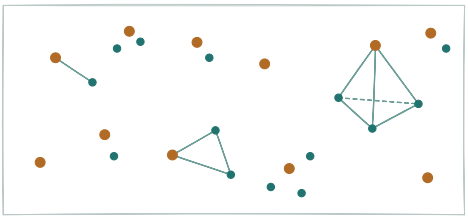
C_z contains only child offsets. **The parent is retained**, giving 1–4 points per cluster.

Construction. Subtract tuples from different parents from the target counts. Then build four-point clusters, fill missing triple and pair counts with smaller clusters, and add singletons to set intensity.

Why small intensity? Different-parent contributions shrink faster than the target densities. Their subtraction can then leave **nonnegative cluster densities**.

Personal feedback

Aspect	Feedback
Mathematical correctness	**Positive**
Research and process documentation	**Positive**
Text writing	**Medium**
Paper structuring	**Negative**
Mathematical explanation	**Medium**



Future of agentic research

What I expect to change

Stronger models can increasingly plan, divide work into tasks, use tools and delegate to subagents.

For a precise mathematical problem, **one initial prompt may increasingly suffice**. This can still mean substantial autonomous work, with no guarantee of success.

As platforms provide more harness functions, we will have **less to build ourselves**.

I also expect explanations to become **less sloppy and more useful for learning**.

Where a custom harness still helps

- Keep the exact target, hypotheses and criteria for success fixed.
- Require separate review and formal verification.
- Keep human checkpoints for direction and interpretation.
- Record results, failed approaches, sources and dependencies.
- Set compute budgets, stopping rules and model choice.
- When writing papers with AI, set your preferences for writing and illustration styles.

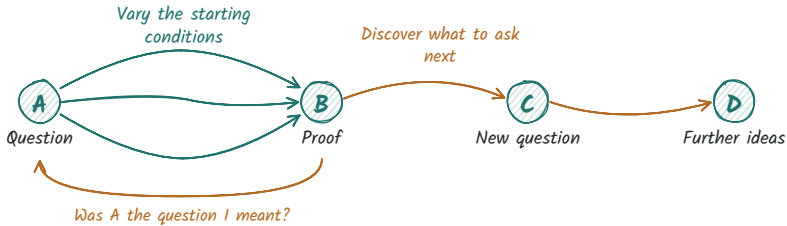
A reported example. For its October release, OpenAI says almost all results came from one prompt to one agent. Some may have needed retries; the full workflow has not been independently reproduced.

Scientific American, October 2026; OpenAI release details.

Personal take

I find this exciting: an elegant solution, with reasoning that feels human.
Still, something is missing when I do not discover the ideas myself.

Faster proofs let me explore more. Research was never a straight line.



Reaching B only helps me if I understand the mathematics.

My concern as a PhD student: outsourcing too much may bypass the learning.
A PhD is still an education. Producing results and developing understanding can come apart.

Related: *A Severe Misalignment of AI in Mathematics* (joint declaration, 11 September 2026).

Thank you!

Questions and discussion

Slides and, later, the experiments:

github.com/MathisRost/agentic-ai-math

